

Prakruthi-BR

BCA (2024) - AI/ML

Subject: Deep Learning

Assignment

Dataset: Banknote Authentication Dataset

it is a classification Dataset taken from UCI Machine learning Repository. it is used to predict whether a banknote is genuine or forged based on numerical features extracted from banknote images. this dataset has 4 input features and 2 output classes.

- Total records = 1372, Input feature = 4, output classes = 2

Type = supervised learning - classification.

Input features:

1. Variance
2. Skewness
3. Kurtosis
4. Entropy

Output features:

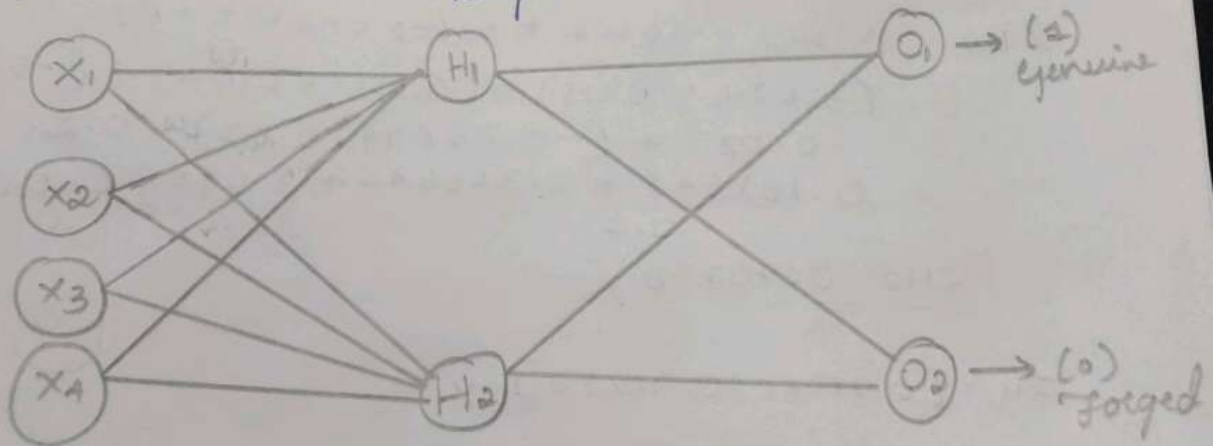
- 0 $\rightarrow$ Genuine
1 $\rightarrow$ Forged.

Neural Network Architecture

Input layer

Hidden layer

Output layer



4 $\longrightarrow$ 2 $\longrightarrow$ 2

All neurons in one layer are connected to all neurons in the next layer. Therefore, this is a fully connected layer.

Weights and Bias:

$4 \times 2 = 8$ weights
↓
Input layer → Hidden layer

$$w_1 = 0.02, w_2 = 0.03, w_3 = 0.04, w_4 = 0.04,$$

$$w_5 = 0.01, w_6 = -0.02, w_7 = 0.03, w_8 = 0.02$$

$$\text{Bias: } B_1 = 0.1, B_2 = 0.2$$

Input: from 1st row,

$$x = \begin{bmatrix} 3.6216 & 8.6661 & -2.8073 & -0.44699 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4$

Output: $y = [1, 0]$ ($1 \rightarrow$ genuine, $0 \rightarrow$ forged)

Hidden layer:

$$\begin{aligned} Z_{H1} &= x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 + B_1 \\ &= (3.6216 \times 0.02) + (8.6661 \times 0.04) + (-2.8073 \times 0.01) + (-0.44699 \times 0.03) + 0.1 \\ &= 0.072432 + 0.086661 - 0.028073 - 0.013410 + 0.1 \end{aligned}$$

$$\boxed{Z_{H1} = 0.21761}$$

$$\begin{aligned} Z_{H2} &= x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + B_2 \\ &= (3.6216 \times 0.03) + (8.6661 \times 0.04) + (-2.8073 \times -0.02) + (-0.44699 \times 0.02) + 0.2 \\ &= 0.108648 + 0.346644 + 0.056146 - 0.008940 + 0.2 \end{aligned}$$

$$\boxed{Z_{H2} = 0.70250}$$

$$Z_H = [0.21761, 0.70250]$$

Activation function:

for hidden layer $\rightarrow$ Sigmoid Activation

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\text{for } H_1 = \frac{1}{1+e^{-0.21761}} = 0.5542$$

$H_1 = 0.5542$

$$\text{for } H_2 = \frac{1}{1+e^{-0.70250}} = 0.6687$$

$H_2 = 0.6687$

$$H = [0.5542, 0.6687]$$

Output Layer

Output weights: $H_1 = w_9 = 0.5$ (genuine(1))
 $w_{11} = -0.4$ (forged(0))
 $H_2 = w_{10} = 0.3$ (genuine(0))
 $w_{12} = -0.2$ (forged(0))

output Bias = $B_3 = 0.1$
 $B_4 = 0.2$

for Genuine (1) =

$$\begin{aligned} Z_{(1)} &= H_1 w_9 + H_2 w_{10} + B_3 \\ &= (0.5542 \times 0.5) + (0.6687 \times 0.3) + 0.1 \\ &= 0.2771 + 0.20061 + 0.1 \\ Z_{(1)} &= 0.57771 \end{aligned}$$

for Forged (0) =

$$\begin{aligned} Z_{(0)} &= H_1 w_{11} + H_2 w_{12} + B_4 \\ &= (0.5542 \times -0.4) + (0.6687 \times -0.2) + 0.2 \\ &= -0.22168 - 0.13374 + 0.2 \\ Z_{(0)} &= -0.15542 \end{aligned}$$

$$Z_{\text{output}} = [0.57771, -0.15542]$$

$$Z_{\text{genuine}} > Z_{\text{forged}}$$

$0.57771 > -0.15542$ = Predicted class = Genuine.

Genuine BankNOTE $\rightarrow$ correct